AERIAL TRACKING & DETECTION SYSTEM

Team Members: Fatih Gazi GÖÇ, Edip Alper SELÇUK, Kaan AYDEMİR
Advisor: Asist. Prof. Dr. Serdar ARSLAN

HAVELSAN SUIT

1 ABSTRACT

This project presents a real-time Edge AI aerial defense system designed to counter the growing security threats posed by unauthorized drones at strategic locations. To overcome the limitations of traditional methods, the system integrates deep learning models with a predictive 'motion prior' tracking algorithm. By analyzing historical movement data, it ensures highly reliable target continuity even on resource-constrained hardware. Ultimately, this research delivers a low-latency, cost-effective airspace security solution adaptable for diverse civilian and military environments.

2 PROBLEM

The increasing use of drones poses critical security threats to strategic infrastructures like military zones and airports. Drones are inherently difficult to detect due to their small size, high agility, and complex backgrounds. While modern transformer-based models offer high accuracy, their computational complexity often prevents real-time operation on edge devices. Furthermore, maintaining object continuity in resource-constrained environments remains a significant challenge. Bridging the gap between high-precision tracking and the strict power limits of edge hardware is essential for developing effective, autonomous ground-to-air defense systems.

3 SOLUTION

Our solution changes the standard tracking pipeline by modifying the order of operations. Instead of using a post-processing filter like DeepSORT or other SORT-based methods, we first predict the next position of the object using a prediction function. This predicted region is then provided to the model as a highlight input before detection. The detection outputs are stored in a tracker list, and this information is used to predict the next position. The process is repeated in a loop, continuously updating the highlight region.

4 METRICS & FORMULAS

Mean Average Precision:

$$\mathrm{mAP} = \frac{1}{N}\sum_{i=1}^{N} AP_i$$

Mean Average Precision 50-95:

$$\mathrm{mAP}_{50-95} = \frac{1}{10}\sum_{t=0.50}^{0.95\ \text{step}\ 0.05} \mathrm{AP}(t)$$

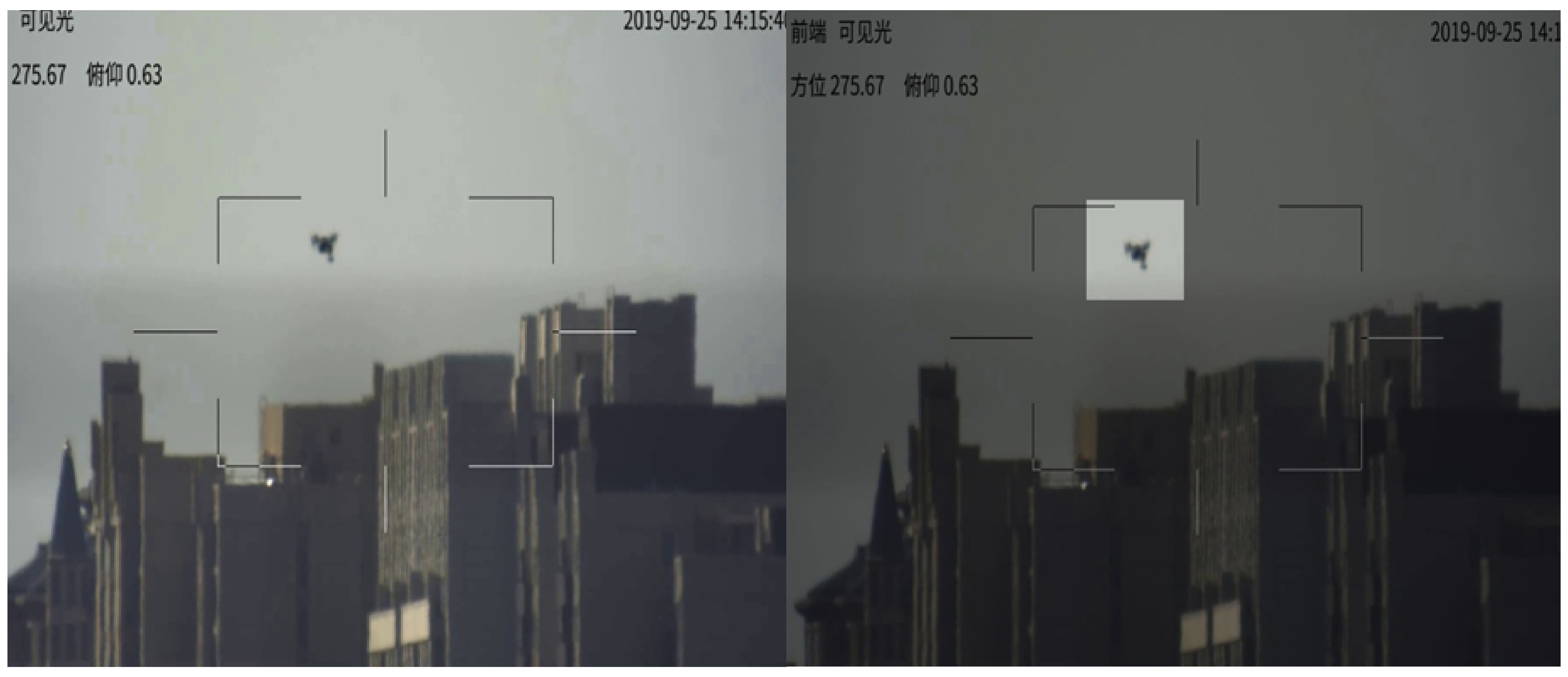
State Accuracy:

$$acc = \sum_{t=1}^{T} \frac{IoU_t \times \delta(v_t > 0) + p_t \times (1-}{T} - 0.2 \times \left(\sum_{t=1}^{T^*} \frac{p_t \times \delta(v_t > 0)}{T^*}\right)^{0.3}$$

Intersection over Union:

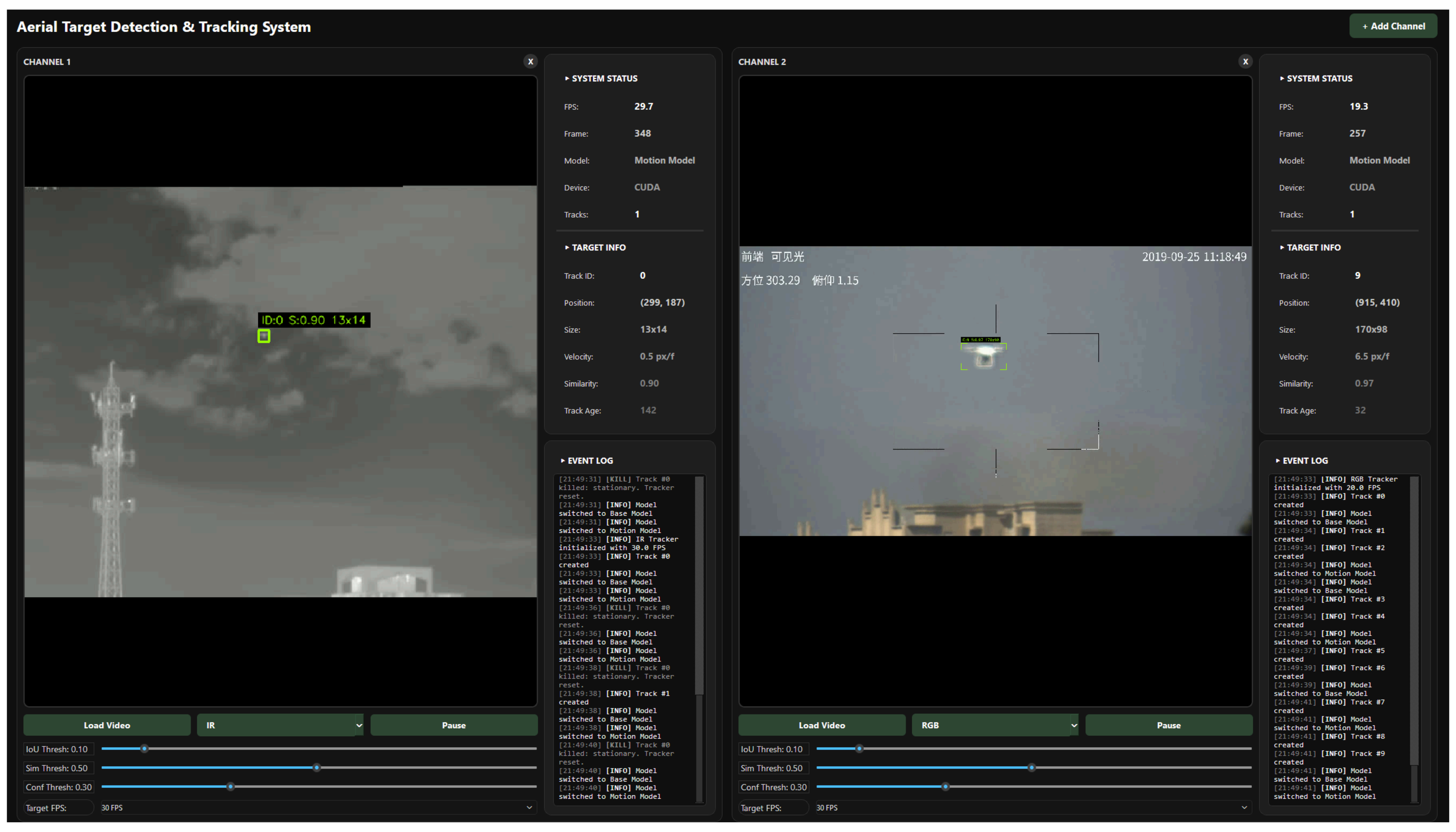
$$IoU = \frac{[\min(x_A^+, x_B^+) - \max(x_A^-, x_B^-)]_+ \, [\min(y_A^+, y_B^+) - \max(y_A^-, y_B^-)]_+}{|A| + |B| - I}$$

5 HIGHLIGHTED IMAGE

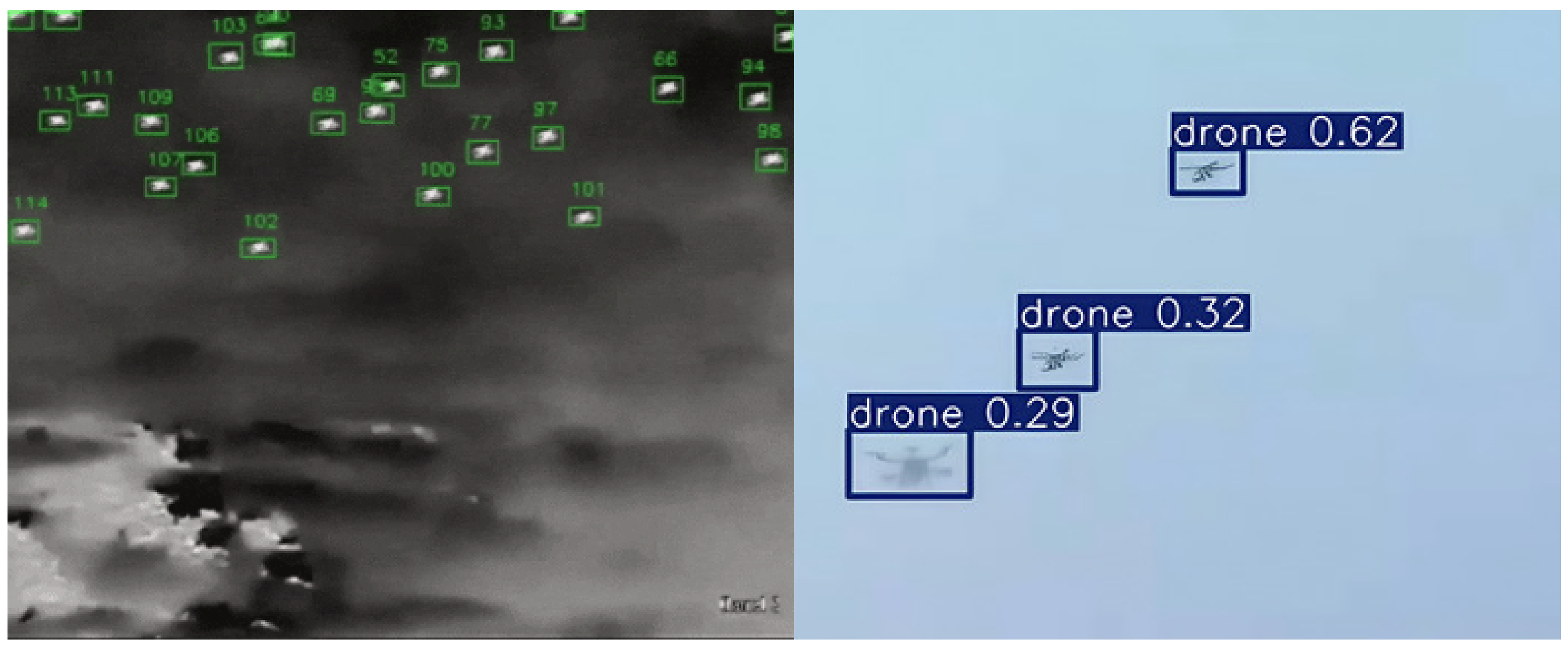


Normal Highlighted

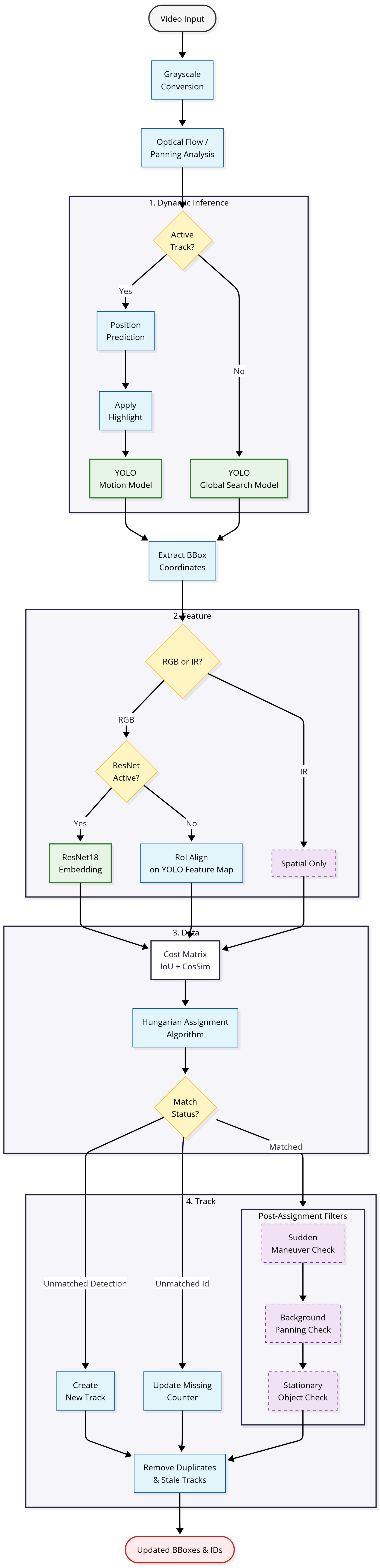
6 USER INTERFACE



7 MULTI DRONE DETECTION & TRACKING



8 SYSTEM ARCHITECTURE



PYQT6 docker NVIDIA CUDA PyTorch OpenCV